Chapter 14. Applications to Group Theory

Covering spaces of graphs, fundamental groups of graphs, subgroups of free groups

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Covering space theory gives proofs of algebraic facts about free groups by realizing free groups as fundamental groups of graphs. (The source PDF does not contain this chapter, so results are named rather than numbered.)

14.1 Covering spaces of a graph

Motivation. As an application of covering-space theory to algebra, we use graphs, whose fundamental groups are free. Covering spaces of graphs are again graphs.

A **linear graph** is a space X that is the union of arcs A_α (**edges**), any two meeting in at most a common end point (a **vertex**), whose topology is coherent with the edges: $C \subset X$ is closed iff $C \cap A_\alpha$ is closed in A_α for every α. A **subgraph** is a union of edges and vertices; it is closed. Fix for each edge a homeomorphism with $[0,1]$. For a subgraph S, the **open star** $\mathrm{St}(S)$ consists of S and the points of edges not in S at parameter distance less than $\frac{1}{2}$ from an end point lying in S; it is open by coherence. We use the standard fact that for compact Hausdorff I, the topology of $X \times I$ is coherent with the subspaces $A_\alpha \times I$ (a product of a quotient map with the identity of a locally compact Hausdorff space is a quotient map).

Lemma 14.1 (graphs are nice)

A linear graph X is Hausdorff and locally path connected; every compact subset meets the interiors of only finitely many edges; and each subgraph S is a deformation retract of $\mathrm{St}(S)$. In particular X is semilocally simply connected.

Proof. Interiors of edges are open in X, and the open stars $\mathrm{St}_\epsilon(v)$ of a vertex (points within parameter distance ϵ of v on the edges at v) are open by coherence; they are path connected, and these sets give enough small neighbourhoods to separate points and to show local path connectedness. If C is compact, pick one point of C in the interior of each edge where possible; every subset of the resulting set D meets each edge in at most one point, so D is closed and discrete in C, hence finite. Pushing each half-open segment of $\mathrm{St}(S) - S$ linearly to its end point in S is continuous on each $(A_\alpha \cap \mathrm{St}(S)) \times I$, hence on $\mathrm{St}(S) \times I$ by coherence, and is a deformation retraction onto S. For a vertex v, $\mathrm{St}(v)$ is contractible; for an interior point of an edge, a small open arc is; so loops in these neighbourhoods are nulhomotopic. ■

Theorem 14.2 (coverings of graphs)

If $p : E \to X$ is a covering map and X a linear graph, then E is a linear graph whose edges are the liftings of the edges of X, and p maps each edge of E homeomorphically onto an edge of X.

Proof. Parametrize an edge A of X by an injective path $f : I \to A$. For each $e \in p^{-1}(f(0))$, the lifting $\tilde{f}$ from e is injective (as $p \circ \tilde{f} = f$ is), so its image is an arc mapped homeomorphically onto A. Each point of $p^{-1}(A)$ lies on exactly one such lifted arc, by uniqueness of path liftings. Lifted arcs over different edges meet at most in points over vertices, which are end points of both; two liftings of the same edge are disjoint. It remains to see that E has the coherent topology. Let $D \subset E$ meet each lifted edge in a closed set, and let $e \in E$. Choose an evenly covered neighbourhood U of $p(e)$ that is a small open star of a vertex or a small open arc in an edge, and the slice $V \ni e$. Each segment of U along one edge is connected, so it lifts into V inside a single lifted edge; thus $p(D \cap V)$ meets each such segment in a closed set, and since U has the topology coherent with these segments, $p(D \cap V)$ is closed in U. Hence $D \cap V$ is closed in V. So each point of $E - D$ has a neighbourhood missing D, and D is closed. ■

14.2 The fundamental group of a graph

Motivation. A connected graph deformation retracts to a wedge of circles after collapsing a maximal tree, so its fundamental group is free, with one generator for each edge outside the tree.

A **tree** is a connected linear graph containing no simple closed curve.

Lemma 14.3 (finite trees)

A finite tree with $n \geq 1$ edges has a vertex lying on only one edge; it deformation retracts onto a vertex; and if it has V vertices then it has $V - 1$ edges.

Proof. Take a maximal sequence of distinct edges forming an arc $e_1 \cup \cdots \cup e_k$ (possible since the tree is finite). Its last end point w lies on no other edge: such an edge would either extend the arc or, meeting the arc again, create a simple closed curve. Removing e_k (except its other end point) leaves a tree T' with $n - 1$ edges and one fewer vertex, and collapsing e_k linearly onto that end point is a deformation retraction of T onto T'. Induct. ■

Theorem 14.4 (trees are simply connected)

Every tree T is simply connected.

Proof. A loop f in T has compact image, which meets the interiors of finitely many edges (Lemma 14.1); let Y be the union of these edges and their end points, and Y_0 the component of Y containing $f(I)$. Y_0 is a finite connected subgraph without simple closed curves, a finite tree, so f is nulhomotopic in $Y_0 \subset T$ (Lemma 14.3). ■

Lemma 14.5 (maximal trees)

Every connected linear graph X contains a maximal tree, and a maximal tree contains every vertex of X.

Proof. The union of a chain of trees (subgraphs) is connected and contains no simple closed curve, since such a curve is compact, lies in finitely many edges and hence in one member of the chain. Zorn's lemma gives a maximal tree T. Suppose a vertex is not in T. If every edge with an end point in T had both end points in T, the subgraph W formed by T's vertices and these edges would be open and closed in X (each edge lies in W or misses it), so $W = X$, a contradiction. So some edge A has exactly one end point in T; $T \cup A$ is a tree (a simple closed curve through the new end point would need two edges there), contradicting maximality. ■

Theorem 14.6 (fundamental group of a graph)

Let X be a connected linear graph, T a maximal tree, x_0 a vertex, and $\{A_\alpha\}$ the edges not in T. For each α let f_α be a loop at x_0 that runs in T to one end of A_α, across A_α, and back in T. Then $\pi_1(X, x_0)$ is free with free generators $\{[f_\alpha]\}$.

Proof.

Step 1. finitely many extra edges, $X = T \cup A_1 \cup \cdots \cup A_n$, by induction on n; $n = 0$ is Theorem 14.4. Let $Y = T \cup A_1 \cup \cdots \cup A_{n-1}$, a an interior point of A_n, p, q the end points of A_n, γ the arc in T from p to q (a finite union of edges), and $C = \gamma \cup A_n$, a simple closed curve. Let $U = X - a$ and $V = \mathrm{St}(C)$ (open stars taken in X). U deformation retracts onto Y (push the two halves of $A_n - a$ to p and q; continuity by the pasting lemma on the closed sets $Y \times I$ and $(A_n - a) \times I$ of

$U \times I$). V deformation retracts onto C, and $U \cap V = V - a$ onto the arc $C - a$ (Lemma 14.1). So $U, V, U \cap V$ are path connected, $U \cap V$ is simply connected, and with base point p, Corollary 11.25 gives $\pi_1(X,p) \cong \pi_1(Y,p) * \pi_1(C,p)$. By induction $\pi_1(Y,p)$ is free on the classes of the loops through $A_1, \dots, A_{n-1}$, and $\pi_1(C,p)$ is infinite cyclic generated by the loop around C; by Theorem 11.19 we get the claim at p, and moving the base point along a path in T transforms these loops into the f_α.

Step 2. general case. Any loop or path homotopy in X has compact image, which meets the interiors of finitely many edges, so it lies in $X_F = T \cup \bigcup_{\alpha \in F} A_\alpha$ for some finite F. Hence the $[f_\alpha]$ generate $\pi_1(X, x_0)$, each has infinite order, and no nonempty reduced word in them represents the identity, since otherwise it would do so already in some X_F, contradicting Step 1. ∎

Corollary 14.7 (Euler characteristic)

If X is a finite connected linear graph with V vertices and E edges, $\pi_1(X)$ is free of rank $E - V + 1 = 1 - \chi(X)$, where $\chi(X) = V - E$.

Proof. A maximal tree contains all V vertices (Lemma 14.5), hence has $V - 1$ edges (Lemma 14.3), leaving $E - V + 1$ edges outside it. ∎

Consequently χ of a finite connected graph depends only on π_1 (by Corollary 11.22), and is unchanged by subdivision.

14.3 Subgroups of free groups

Motivation. Every subgroup of a free group is free (Nielsen–Schreier). The algebraic proof is intricate, but topologically it is immediate: a subgroup of π_1 of a graph is the fundamental group of a covering space, which is again a graph.

A wedge of circles is a linear graph after subdividing each circle into three arcs (the topology, coherent with the circles, is coherent with the arcs). By Lemma 11.30 and Theorem 11.29, every free group is the fundamental group of such a graph.

Theorem 14.8 (Nielsen–Schreier)

Every subgroup H of a free group F is free.

Proof. Let X be a wedge of circles with $\pi_1(X, x_0) \cong F$. X is path connected, locally path connected and semilocally simply connected (Lemma 14.1), so Theorem 13.15 gives a covering $p : E \to X$ with $p_*\pi_1(E, e_0) = H$. As p_* is injective, $H \cong \pi_1(E, e_0)$. E is a connected linear graph (Theorem 14.2), so $\pi_1(E, e_0)$ is free (Theorem 14.6). ∎

Theorem 14.9 (Schreier index formula)

If F is free of rank n and H has finite index k in F, then H is free of rank $k(n-1)+1$.

Proof. In the proof above, the fibre $p^{-1}(x_0)$ is in bijection with the right cosets of H (Theorem 9.17), so p is k-sheeted: each vertex and each edge of X has exactly k preimages. The subdivided wedge X has $2n+1$ vertices and $3n$ edges, so $\chi(X) = 1 - n$ and $\chi(E) = k(1-n)$. By Corollary 14.7, $\pi_1(E)$ has rank $1 - \chi(E) = k(n-1)+1$. ∎

Example 14.1

(a) Let F be free on a, b, the fundamental group of the figure eight X. The grid of horizontal and vertical lines through integer points of $\mathbb{R}^2$ covers X (Example 9.3); its covering transformations are the integer translations, acting transitively on fibres, so the covering is regular with $F/H \cong \mathbb{Z}^2$, $H = p_*\pi_1(E)$. Since F/H is abelian, $H \supset [F, F]$; the induced surjection $\mathbb{Z}^2 \cong F/[F, F] \to F/H \cong \mathbb{Z}^2$ is injective (a surjective endomorphism of $\mathbb{Z}^2$ has determinant ± 1), so $H = [F, F]$. The grid has the comb (the x-axis and all vertical lines) as a maximal tree, leaving infinitely many edges, so $[F, F]$ is free of infinite rank.
(b) The kernel of $F \to \mathbb{Z}/2$, $a, b \mapsto 1$, has index 2, so it is free of rank $2(2-1)+1 = 3$. Thus, unlike for free abelian groups, a subgroup of a free group can have larger rank than the group.

14.4 Selected exercises

Motivation. The source has no exercises for this chapter, so the following are standard consequences of the theorems above, with our own solutions: trees are recognized by counting edges, free groups have no torsion, and the Schreier formula produces free subgroups of any finite rank and counts the subgroups of index 2.

Exercise 14.1 (recognizing finite trees)

Let X be a finite connected linear graph with V vertices and E edges.
(a) Show that $E \geq V - 1$, with equality iff X is a tree.
(b) Show that X is a tree iff $\pi_1(X)$ is trivial.

Solution [Supplement].

(a) Let T be a maximal tree in X (Lemma 14.5). It contains all V vertices, so it has $V - 1$ edges (Lemma 14.3; if T has no edges, then $V = 1$). Hence $E \geq V - 1$. If X is a tree, then $T = X$ and $E = V - 1$. If $E = V - 1$, then every edge of X lies in T; since X is the union of its edges and vertices, $X = T$ is a tree.

(b) A tree is simply connected (Theorem 14.4). Conversely, if $\pi_1(X)$ is trivial, then by Corollary 14.7 it is free of rank $E - V + 1 = 0$, so $E = V - 1$ and X is a tree by (a). ■

Exercise 14.2 (free groups are torsion free)

Show that every element $g \neq e$ of a free group F has infinite order.

Solution [Supplement].

Step 1. (A free group on a nonempty set is infinite.) Let H be free on a nonempty set S and $s \in S$. By Lemma 11.18 there is a homomorphism $\phi : H \to \mathbb{Z}$ with $\phi(s) = 1$ and $\phi(s') = 0$ for $s' \neq s$. Then $\phi(s^k) = k$, so the powers s^k, $k \in \mathbb{Z}$, are distinct.

Step 2. Suppose $g \neq e$ has finite order. Then $\langle g \rangle$ is a finite group with more than one element. By the Nielsen–Schreier theorem (Theorem 14.8) it is free on some set S, and $S \neq \emptyset$ because $\langle g \rangle$ is nontrivial. This contradicts Step 1. ■

Exercise 14.3 (free subgroups of every finite rank)

Let F be free on $\{a, b\}$. Show that for every $n \geq 1$, F has a subgroup that is free of rank n.

Solution [Supplement].

Step 1. ($n = 1$.) By Step 1 of Exercise 14.2, a has infinite order, so $\langle a \rangle \cong \mathbb{Z}$ is free on $\{a\}$.

Step 2. ($n \geq 2$.) Let $k = n - 1 \geq 1$. By Lemma 11.18 there is a homomorphism $\phi : F \to \mathbb{Z}/k$ with $\phi(a) = 1$ and $\phi(b) = 0$. It is onto, so $H = \ker \phi$ has $F/H \cong \mathbb{Z}/k$, i.e. index k.

Step 3. By the Schreier index formula (Theorem 14.9), H is free of rank $k(2-1)+1 = k+1 = n$. ■

Exercise 14.4 (subgroups of index 2)

Let F be free of rank $n \geq 1$. Show that F has exactly $2^n - 1$ subgroups of index 2, and that each is free of rank $2n - 1$.

Solution [Supplement].

Step 1. A subgroup H of index 2 is normal: for $g \notin H$ both the left coset gH and the right coset Hg equal $F - H$. So F/H is a group of order 2, and $H = \ker \phi$ for the surjective homomorphism $\phi : F \to F/H \cong \mathbb{Z}/2$. Conversely, the kernel of a surjection $F \to \mathbb{Z}/2$ has index 2.

Step 2. A surjection $\phi : F \to \mathbb{Z}/2$ is determined by its kernel H: $\phi(x) = 0$ for $x \in H$ and $\phi(x) = 1$ otherwise. So the subgroups of index 2 correspond bijectively to the surjections $F \to \mathbb{Z}/2$.

Step 3. Let $x_1, \ldots, x_n$ be free generators. By Lemma 11.18, homomorphisms $F \to \mathbb{Z}/2$ correspond bijectively to maps $\{x_1, \ldots, x_n\} \to \mathbb{Z}/2$, of which there are 2^n. Only the zero map gives a homomorphism that is not onto, so there are $2^n - 1$ surjections, and $2^n - 1$ subgroups of index 2.

Step 4. By Theorem 14.9, each has rank $2(n-1)+1 = 2n-1$. ■